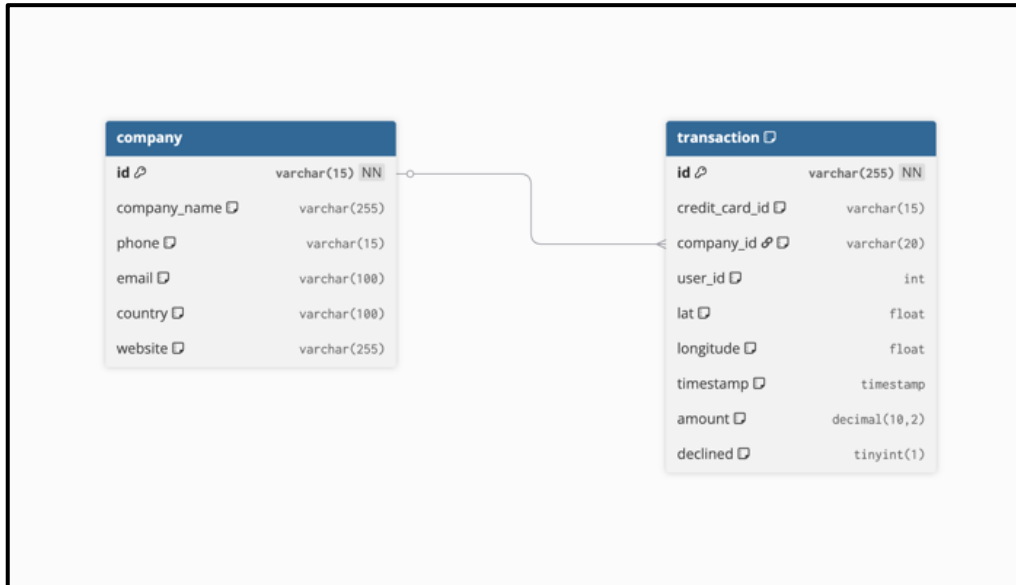


NIVELL 1

EXERCICI 1



Las dos tablas se relacionan en una relación de 1 a varios. La tabla **Company** tiene como PK el campo **id**, la tabla **Transaction** tiene como PK el campo **id**, pero este no es el mismo que en la tabla **Company**. **Company** se relaciona con **Transaction** con la FK **company_id** que correspondería con la **id** de **Company**.

EXERCICI 2

- Llistat dels països que estan generant vendes.

```

7 • SELECT DISTINCT(c.country)
8   FROM company AS c
9   JOIN transaction AS t ON t.company_id = c.id
10  WHERE t.declined = 0;
11
12  #Des de quants països es generen les vendes.#
13
14 • SELECT COUNT(DISTINCT(country)) AS num_paises
15   FROM company
16   JOIN transaction ON transaction.company_id = company.id
17   WHERE declined = 0;
    
```

Result Grid

country
United...
Ireland
United...
Belgium
Canada
Germa...
Norway
France
Italy
United...

Result 65

Action Output

	Time	Action	Response	Duration / Fetch Time
1	12:59:13	SELECT DISTINCT(c.country) FROM company AS c JOIN transaction AS t ON t.company_id = c.id...	15 row(s) returned	0.254 sec / 0.00000...

- Des de quants països es generen les vendes.

```

14 • SELECT COUNT(DISTINCT(c.country)) AS num_paises
15 FROM company AS c
16 JOIN transaction AS t ON t.company_id = c.id
17 WHERE t.declined = 0;
18
19 #Identifica la companyia amb la mitjana més gran de vendes.#
20
21 • SELECT c.company_name,
22         ROUND(AVG(t.amount),2) as vendes_mitjana
23 FROM company AS c
24 JOIN transaction AS t ON t.company_id = c.id
25 WHERE t.declined = 0;
26
27 #Exercici_3#

```

Result Grid

num_paises
15

Result 66

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	13:01:51	SELECT COUNT(DISTINCT(c.country)) AS num_paises FROM company AS c JOIN transaction AS t...	1 row(s) returned	0.254 sec / 0.00000...

- Identifica la companyia amb la mitjana més gran de vendes.

```

21 • SELECT c.company_name,
22         ROUND(AVG(t.amount),2) as vendes_mitjana
23 FROM company AS c
24 JOIN transaction AS t ON t.company_id = c.id
25 WHERE t.declined = 0
26 GROUP BY c.company_name
27 ORDER BY vendes_mitjana DESC LIMIT 1;
28
29 #Exercici_3#

```

Result Grid

company_name	vendes_mitjana
Ac Fermentum Incorporated	284.91

Result 62

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	12:56:58	SELECT c.company_name, ROUND(AVG(t.amount),2) as vendes_mitjana FROM company AS c JO...	1 row(s) returned	0.325 sec / 0.000007...

EXERCICI 3

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
33 SELECT t.*
34 FROM transaction AS t
35 WHERE t.declined = 0 AND t.company_id IN (SELECT c.id
36 FROM company AS c
37 WHERE c.country = 'Germany');
38
39 #Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.#
40
41 SELECT company_name
```

id	credit_card...	company_id	user_id	lat	longitude	timestamp	amount	declined
00001A11-2E82-44CA-941D-198FC8D077F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
00687139-4B82-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4869	2019-03-09 19:37:49	524.84	0
0074FADD-3DF1-4827-8758-55898314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0
00AA8BCD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-8797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0
00BE9D4-6920-47D8-ABE8-325E2269829D	CcS-4983	b-2222	402	38.708	-9.12993	2019-02-27 15:25:16	141.66	0
00DA0383-E048-4577-8ED1-3C56C258FF2F	CcS-9223	b-2222	4642	51.1742	10.2027	2019-03-21 11:47:34	325.62	0
00DD11DE-ED01-4BBD-93AD-174D183A59DF	CcS-7881	b-2222	3100	45.7565	4.83109	2024-01-28 18:20:49	242.53	0

Time	Action	Response	Duration / Fetch Time
12:51:30	SELECT * FROM transaction AS t WHERE t.declined = 0 AND t.company_id IN (SELECT c.id...	13269 row(s) returned	0.0012 sec / 0.054 sec

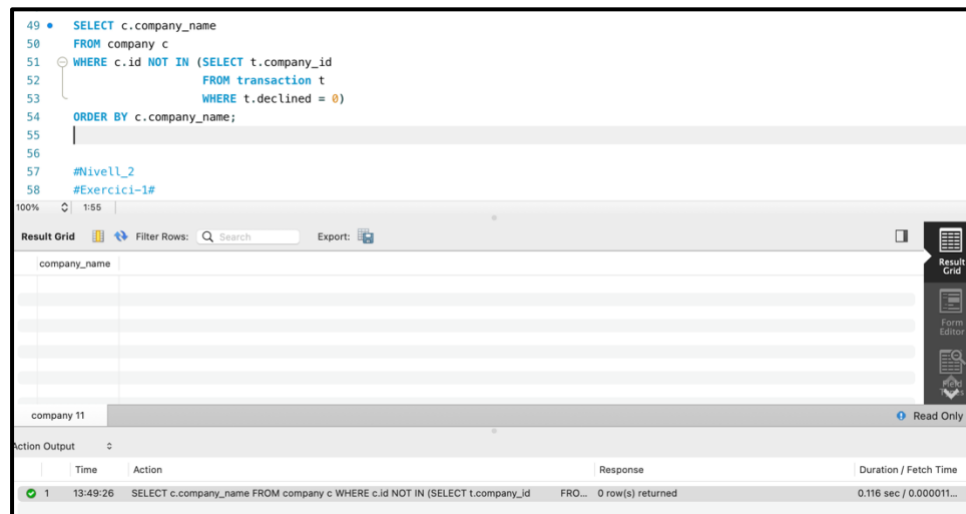
- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
36 SELECT c.company_name
37 FROM company c
38 WHERE c.id IN (SELECT t.company_id
39 FROM transaction t
40 WHERE t.declined = 0 AND t.amount > (SELECT AVG(t.amount)
41 FROM transaction t
42 WHERE t.declined = 0
43 ));
44
45
46
47 #Eliminar del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.#
48
```

company_name
Fuseco Corp.
Convalis In Incorporated
Ante laeulis Nec Foundation
Donec Ltd
Sed Nunc Ltd
Amet Nulla Donec Corporation
Nascentur Ridiculus Mus Inc.
Vestibulum Lorem PC
Gravida Sagittis LLP
Aliquam Amet Egestas Foundation

Time	Action	Response	Duration / Fetch Time
13:45:25	SELECT c.company_name FROM company c WHERE c.id IN (SELECT t.company_id FROM...	100 row(s) returned	0.193 sec / 0.000051...

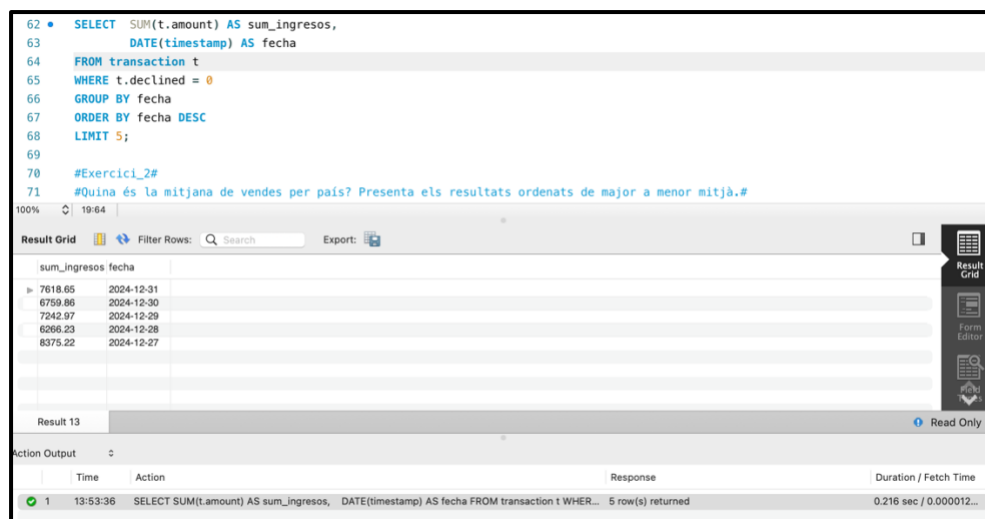
- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.



Nivell 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.



Exercici 2

Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
73 SELECT ROUND(AVG(t.amount), 2) AS mitj_vendes,
74 c.country
75 FROM transaction t
76 JOIN company c ON c.id = t.company_id
77 WHERE t.declined = 0
78 GROUP BY c.country
79 ORDER BY mitj_vendes DESC;
80
81 #Exercici_3#
```

mitj_vendes	country
260.97	Belgium
260.83	Germany
260.39	Ireland
260.28	Spain
259.91	France
259.59	New Zealand
259.14	Norway
258.34	Netherlands
258.12	Italy
257.44	Sweden

Action Output

	Time	Action	Response	Duration / Fetch Time
1	14:01:55	SELECT ROUND(AVG(t.amount), 2) AS mitj_vendes, c.country FROM transaction t JOIN company...	15 row(s) returned	0.737 sec / 0.000014...

Exercici 3

Demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

- Mostra el llistat aplicant JOIN i subconsultes.

```
85 SELECT t.*
86 FROM transaction t
87 JOIN company c ON c.id = t.company_id
88 WHERE t.declined = 0 AND c.country = (SELECT c.country
89 FROM company c
90 WHERE c.company_name = "Non Institute");
91
92 #Mostra el llistat aplicant solament subconsultes.#
93
94 SELECT transaction.id
```

id	credit_card...	company_id	user_id	lat	longitude	timestamp	amount	declined
01F075B1-D7AE-4D02-AA9D-5FFD72A43F3C	CcS-8700	b-2246	4119	55.856	-3.15783	2018-01-27 13:44:36	103.73	0
023FFCEB-E618-4938-BF56-C8DF90540ADD	CcS-7816	b-2246	3235	46.3568	1.82755	2016-12-19 11:53:45	219.28	0
028338EB-EF91-4564-957B-D6F1662AB7C5	CcS-9471	b-2246	4890	42.1332	12.396	2017-01-10 21:09:29	328.87	0
02CF296C-CEF2-4C1E-A594-F478BF279C0	CcS-9082	b-2246	4501	39.4662	-0.373246	2020-05-24 01:17:29	155.72	0
02F468DC-42EC-47C2-8B9A-D8B2587A81AF	CcS-6913	b-2246	2332	52.175	19.3508	2023-03-17 16:36:27	305.35	0
0306BE3B-817B-4A49-934E-0E439291A104	CcS-5302	b-2246	721	51.9233	18.926	2021-12-02 23:06:02	339.58	0
0347BF6E-9EB5-4387-B187-0E78E8F288FB	CcS-7674	b-2246	3093	45.768	4.84271	2021-12-30 08:40:24	172.93	0
03AEB0DE-D097-4B03-9C57-6A6D878026FD	CcS-6121	b-2246	1540	50.8113	10.3145	2018-11-11 11:28:49	114.77	0
03CA38D3-88FF-4DBF-BFD4-4CC7DAEED2B	CcS-8036	b-2246	3455	52.5178	13.4131	2017-02-25 15:38:21	440.27	0
044448B8-8B9D-43ED-80CE-ED0CE31A637	CcS-6704	b-2246	2310	41.0643	12.4657	2018-05-17 13:53:53	241.63	0

Action Output

	Time	Action	Response	Duration / Fetch Time
1	14:00:18	SELECT t.* FROM transaction t JOIN company c ON c.id = t.company_id WHERE t.declined = 0 AND...	13754 row(s) returned	0.0033 sec / 0.090 sec

- Mostra el llistat aplicant solament subconsultes.

The screenshot shows a SQL query in a query editor. The query is as follows:

```

94 SELECT t.*
95 FROM transaction t
96 WHERE t.declined = 0 AND t.company_id IN (SELECT c.id
97 FROM company c
98 WHERE c.country IN (SELECT c.country
99 FROM company c
100 WHERE c.company_name = "Non Institute"))
101 );
102
103
104 #nivell_3
100% 1:102

```

The result grid shows the following data:

id	credit_card...	company_id	user_id	lat	longitude	timestamp	amount	declined
001ESD06-A391-4735-88D8-748F16C061A6	CcS-7425	b-2310	2844	59.9165	18.5518	2021-09-10 02:39:32	418.20	0
0022377F-4447-432B-801A-CFE5416E336C	CcS-8582	b-2522	4001	50.7865	10.6173	2019-10-24 06:52:23	171.13	0
00285171-68B7-4E96-9787-BD580BE4515D	CcS-8358	b-2310	3777	55.7751	-3.8232	2022-09-21 21:18:51	192.60	0
0031C741-BE1F-43F3-954F-4F3389953F54	CcS-6854	b-2374	2273	51.6159	19.069	2015-09-22 04:36:39	381.37	0
0031FD57-439C-4412-8ABB-51812F0BD2AD	CcS-6023	b-2522	1442	59.7896	18.7997	2022-08-27 23:04:07	118.22	0
003B333E-DBCC-403C-8B0A-B5903E3E7AA5	CcS-5129	b-2310	548	41.394	2.16367	2019-01-05 22:10:32	26.66	0
0038A409-A589-4B6C-A87D-30C0C232124	CcS-6255	b-2374	1674	39.7493	-8.18479	2018-04-01 01:39:23	318.72	0
003AD3F7-AF0F-441C-B221-6117814CA2F7	CcS-6228	b-2618	1647	55.3732	-3.39702	2021-01-14 20:39:57	266.68	0
003E193D-81AB-4097-ACAD-A5D221D4DE24	CcS-6224	b-2326	1643	42.1974	12.7358	2017-11-02 15:45:57	130.56	0
003E555E-1E0D-433A-89E7-F76F77D90EC8	CcS-5442	b-2430	861	52.3788	19.6379	2022-06-04 01:06:21	12.45	0
0041CB27-67E3-4047-8A9E-87A37A64CFDC	CcS-8824	b-2522	4243	51.408	10.3739	2018-06-17 13:21:42	209.23	0
00457D41-C732-449A-87CD-84D9935C1946	CcS-5994	b-2310	1413	51.816	19.8381	2022-09-03 02:47:05	386.35	0
0045F54B-D4D1-4582-9C3C-F1DA4A4A40D5	CcS-7661	b-2574	3080	48.8648	2.34847	2019-02-07 03:27:04	269.00	0
00472998-74CA-48FD-9606-6373516777BF	CcS-7634	b-2522	3053	45.7724	4.82578	2020-09-15 21:02:14	530.53	0

The action output shows the following results:

Time	Action	Response	Duration / Fetch Time
14:07:01	SELECT t.* FROM transaction t WHERE t.declined = 0 AND t.company_id IN (SELECT c.id...	13754 row(s) returned	0.0089 sec / 0.897 sec

Nivell 3

Exercici 1

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

The screenshot shows a SQL query in a query editor. The query is as follows:

```

94 SELECT t.*
95 FROM transaction t
96 WHERE t.declined = 0 AND t.company_id IN (SELECT c.id
97 FROM company c
98 WHERE c.country IN (SELECT c.country
99 FROM company c
100 WHERE c.company_name = "Non Institute"))
101 );
102
103
104 #nivell_3
100% 1:102

```

The result grid shows the following data:

id	credit_card...	company_id	user_id	lat	longitude	timestamp	amount	declined
001ESD06-A391-4735-88D8-748F16C061A6	CcS-7425	b-2310	2844	59.9165	18.5518	2021-09-10 02:39:32	418.20	0
0022377F-4447-432B-801A-CFE5416E336C	CcS-8582	b-2522	4001	50.7865	10.6173	2019-10-24 06:52:23	171.13	0
00285171-68B7-4E96-9787-BD580BE4515D	CcS-8358	b-2310	3777	55.7751	-3.8232	2022-09-21 21:18:51	192.60	0
0031C741-BE1F-43F3-954F-4F3389953F54	CcS-6854	b-2374	2273	51.6159	19.069	2015-09-22 04:36:39	381.37	0
0031FD57-439C-4412-8ABB-51812F0BD2AD	CcS-6023	b-2522	1442	59.7896	18.7997	2022-08-27 23:04:07	118.22	0
003B333E-DBCC-403C-8B0A-B5903E3E7AA5	CcS-5129	b-2310	548	41.394	2.16367	2019-01-05 22:10:32	26.66	0
0038A409-A589-4B6C-A87D-30C0C232124	CcS-6255	b-2374	1674	39.7493	-8.18479	2018-04-01 01:39:23	318.72	0
003AD3F7-AF0F-441C-B221-6117814CA2F7	CcS-6228	b-2618	1647	55.3732	-3.39702	2021-01-14 20:39:57	266.68	0
003E193D-81AB-4097-ACAD-A5D221D4DE24	CcS-6224	b-2326	1643	42.1974	12.7358	2017-11-02 15:45:57	130.56	0
003E555E-1E0D-433A-89E7-F76F77D90EC8	CcS-5442	b-2430	861	52.3788	19.6379	2022-06-04 01:06:21	12.45	0
0041CB27-67E3-4047-8A9E-87A37A64CFDC	CcS-8824	b-2522	4243	51.408	10.3739	2018-06-17 13:21:42	209.23	0
00457D41-C732-449A-87CD-84D9935C1946	CcS-5994	b-2310	1413	51.816	19.8381	2022-09-03 02:47:05	386.35	0
0045F54B-D4D1-4582-9C3C-F1DA4A4A40D5	CcS-7661	b-2574	3080	48.8648	2.34847	2019-02-07 03:27:04	269.00	0
00472998-74CA-48FD-9606-6373516777BF	CcS-7634	b-2522	3053	45.7724	4.82578	2020-09-15 21:02:14	530.53	0

The action output shows the following results:

Time	Action	Response	Duration / Fetch Time
14:07:01	SELECT t.* FROM transaction t WHERE t.declined = 0 AND t.company_id IN (SELECT c.id...	13754 row(s) returned	0.0089 sec / 0.897 sec

Exercici 2

Demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen més de 400 transaccions o menys.

123
124 • `SELECT COUNT(t.id) AS num_trans,`
125 `c.company_name,`
126
127 `CASE`
128 `WHEN COUNT(t.id) > 400 THEN 'The quantity of transactions is greater than 400'`
129 `ELSE 'The quantity of transactions is under 400'`
130 `END`
131 `FROM transaction t`
132 `JOIN company c ON c.id = t.company_id`
133 `WHERE t.declined = 0`
134 `GROUP BY c.company_name`
`ORDER BY num_trans DESC;`

100% 19:130

Result Grid

num_trans	company_name	CASE
1590	Donec Fingilla PC	The quantity of transactions is greater than 400
1583	Mauris Institute	The quantity of transactions is greater than 400
1581	Aliquet Vel Vulputate Incorporated	The quantity of transactions is greater than 400
1578	Rutrum Non Inc.	The quantity of transactions is greater than 400
1573	Aliquam PC	The quantity of transactions is greater than 400
1565	Orci Adipiscing Limited	The quantity of transactions is greater than 400
1564	Eliam Bibendum Fermentum Industries	The quantity of transactions is greater than 400
1563	Mus Aenean Eget Foundation	The quantity of transactions is greater than 400

Result 3

Action Output

	Time	Action	Response	Duration / Fetch Time
1	15:31:24	SELECT COUNT(t.id) AS num_trans, c.company_name, CASE WHEN COUNT(t.id) > 400 THEN...	100 row(s) returned	0.813 sec / 0.000031...